Bichemistry Cume
11-09-07

1. A peptide, AVARTYS was digested with trypsin (which cleaves a peptide bond at carboxyl end of Lys and Arg residues).
   A. What are the resulting peptides?
   B. What are the pI values of the resulting peptides (use the pKa values listed below).
   C. The CM-column was used to separate the peptides resulted from the trypsin digest. Predict the order of elution using an increasing concentration NaCl gradient.
   D. Draw the chemical structure of the smallest fragment resulted from the trypsin digest. Show all the hydrogen atoms expected to be present at pH 7.

2. Many proteins are known to bind metals. X-ray diffraction method is the only method to directly determine metal-protein interactions. However, X-ray diffraction method may not be able to tell apart Ca2+ atom from K+? Why?

Amino Acid pKa values

Cys	8.6
Asp	4.5
Glu	4.5
His	6.8
Lys	11.1
Arg	12.5
Tyr	9.8

Amine	9.5
Carboxylic	2.3